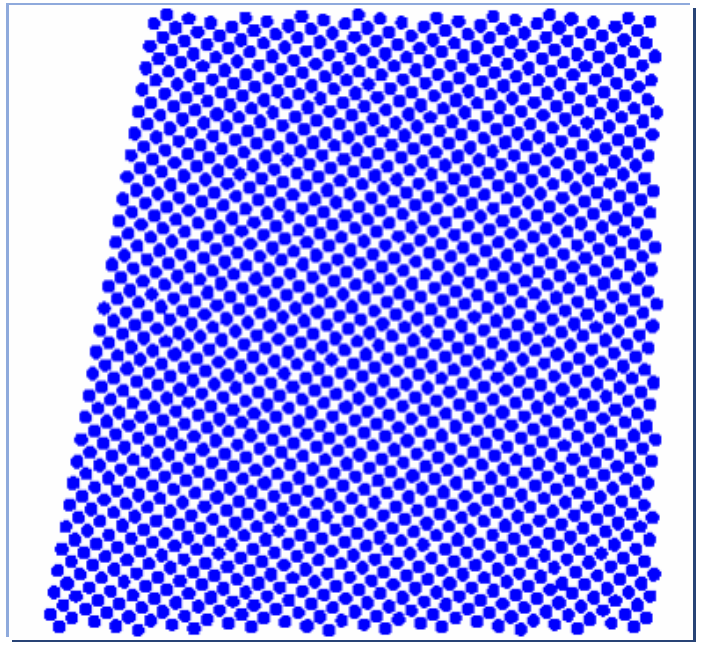
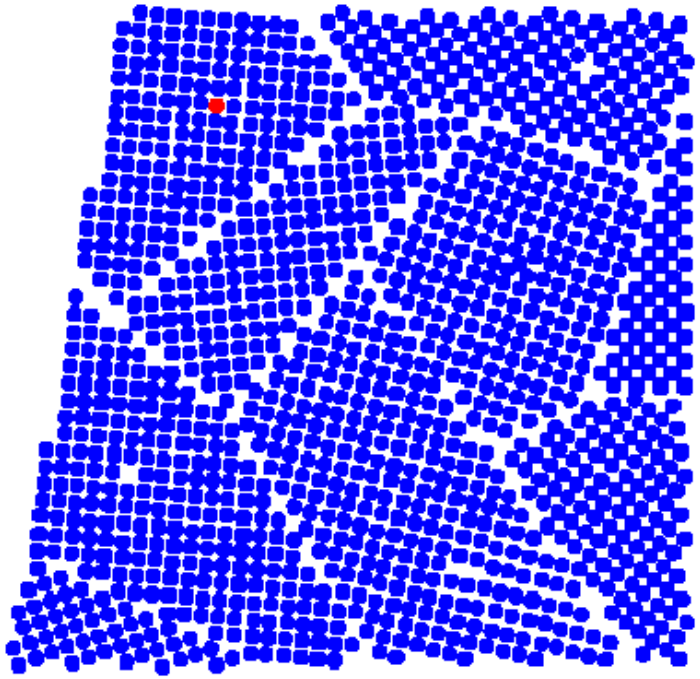
ME 282

REAL CRYSTALS

- **Real crystals** are never perfect: there are **always defects**!



Perfect crystal



Real crystal

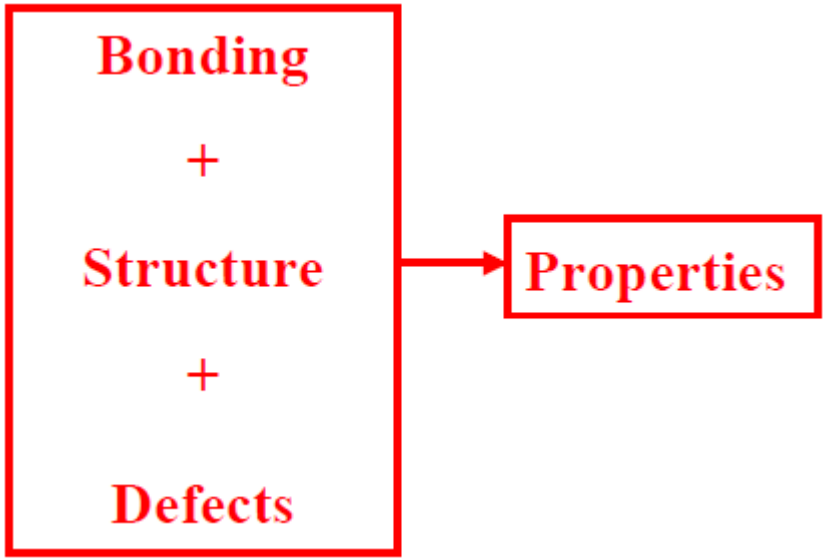
- **Controlling** the defects is one of **the main** goal of an engineer!

Schematic drawing of a poly-crystal with many defects by Helmut Föll, University of Kiel, Germany.

Imperfections in Solids

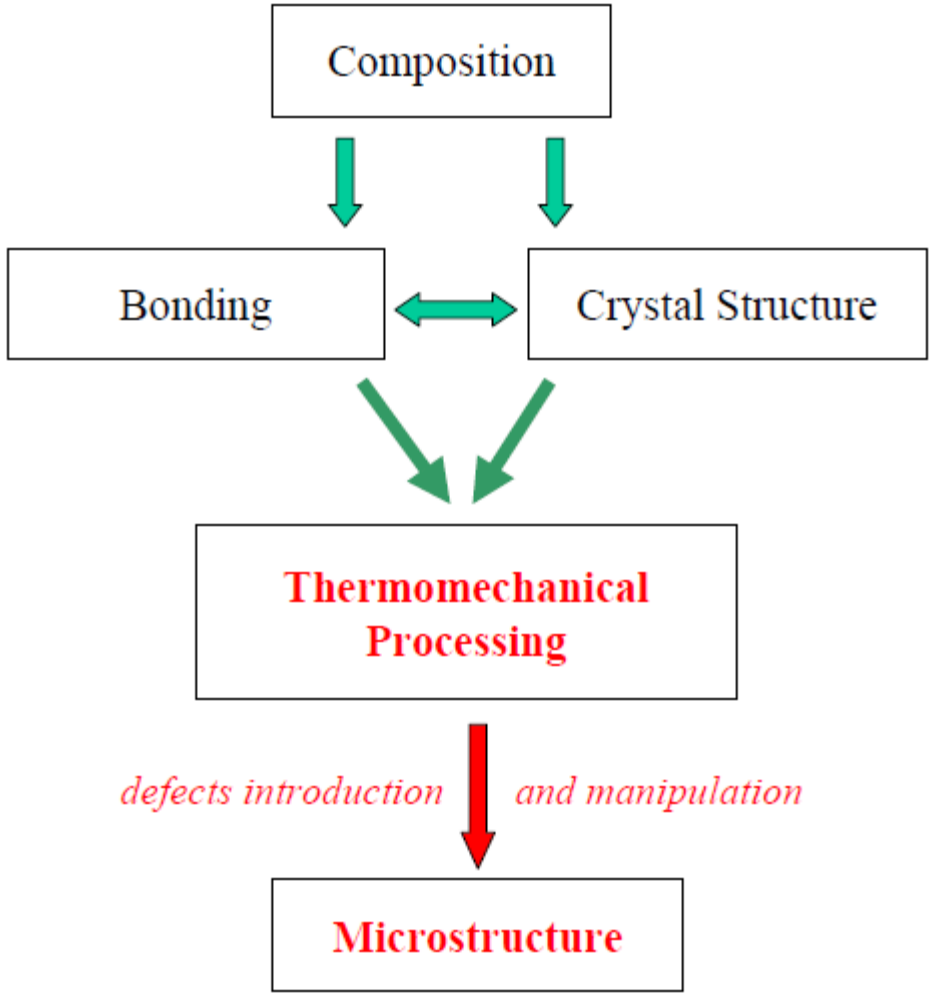
- The properties of materials are profoundly influenced by the presence of imperfections.
- It is important to have knowledge about the types of imperfections that exist and the roles they play in affecting the behavior of materials.
- Defects in the crystalline structure can have a tremendous effect on macroscopic materials properties

- We can modify and improve many of the physical, electrical, magnetic, and optical properties of crystalline materials by controlling the imperfections in their lattice structure.



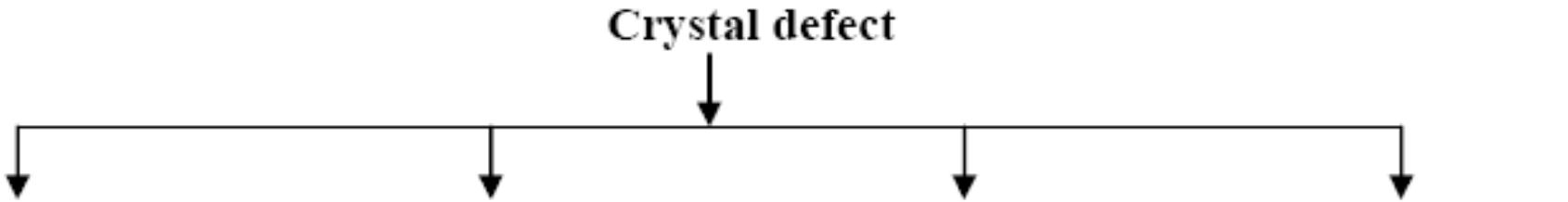
Defects Introduction and Manipulation

- The processing determines the defects



Types of Defects

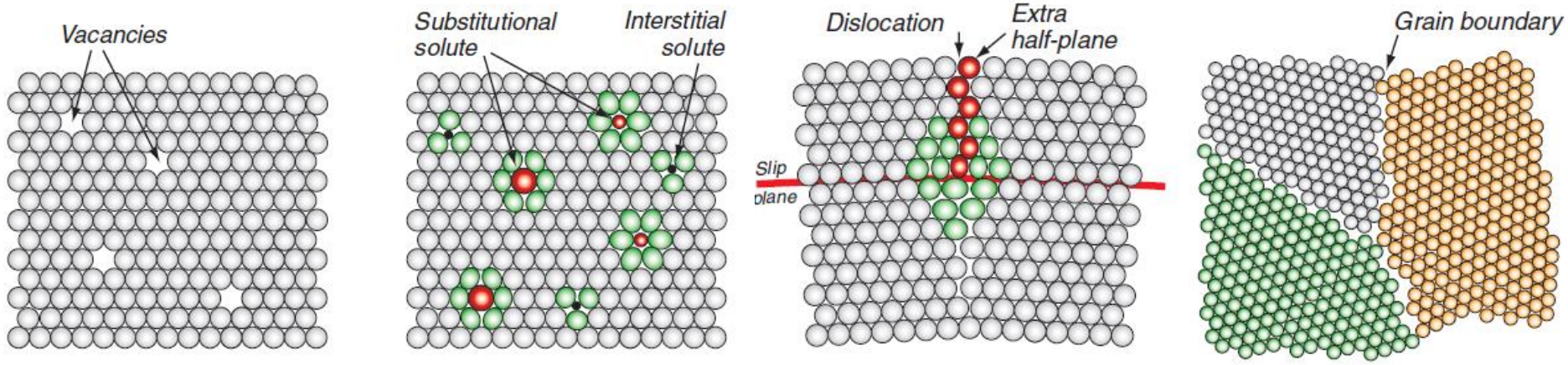
- Defects may be classified into four categories depending on their dimension:



Point defects	Line defects	Surface defects	Volume defects
Vacancy	Edge dislocation	Grain boundaries	Inclusions
Schottkey	Screw Dislocation	Titl boundaries	Voids
Self interstitial		Twin boundaries	
Frenkel		Stacking faults	
Substitutional			

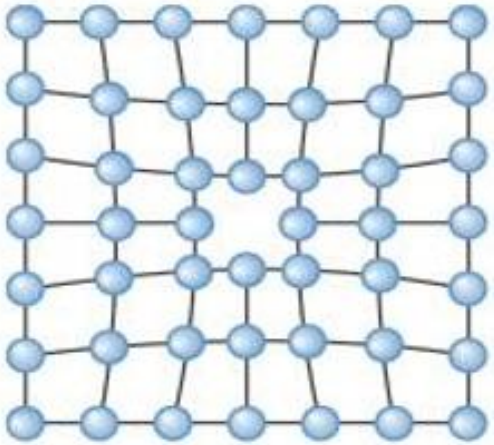
Types of Defects

- 0D, Point defects: atoms missing or in irregular places in the lattice (vacancies, interstitials and Substitutional)
- 1D, Linear defects: groups of atoms in irregular positions (e.g. screw and edge dislocations)
- 2D, Planar defects: the interfaces between homogeneous regions of the material (grain boundaries, external surfaces)
- 3D, Volume defects: extended defects (pores, cracks)



Vacancy Defects

- A vacancy defect is formed when an atom or an ion is missing from a normal atomic site in a crystalline structure.
- The number of vacancies present in a material increases exponentially as the temperature increases.



(a) vacancy

- The equilibrium number of vacancies formed as a result of thermal vibrations may be calculated from thermodynamics:

$$N_v = N_s \exp\left(-Q_v / k_B T\right)$$

Note

$$N_s = \frac{N_A \rho}{A}$$

where N_s is the number of regular lattice sites, k_B is the Boltzmann constant, Qv is the energy needed to form a vacant lattice site in a perfect crystal, and T the temperature in Kelvin (note, not in ^oC or ^oF).

Measuring Activation Energy

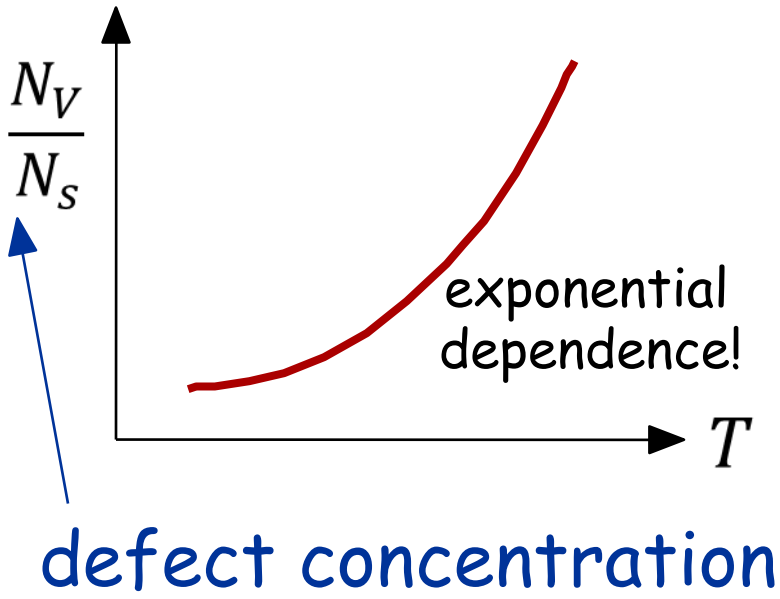
- We can get Q_v from an experiment.

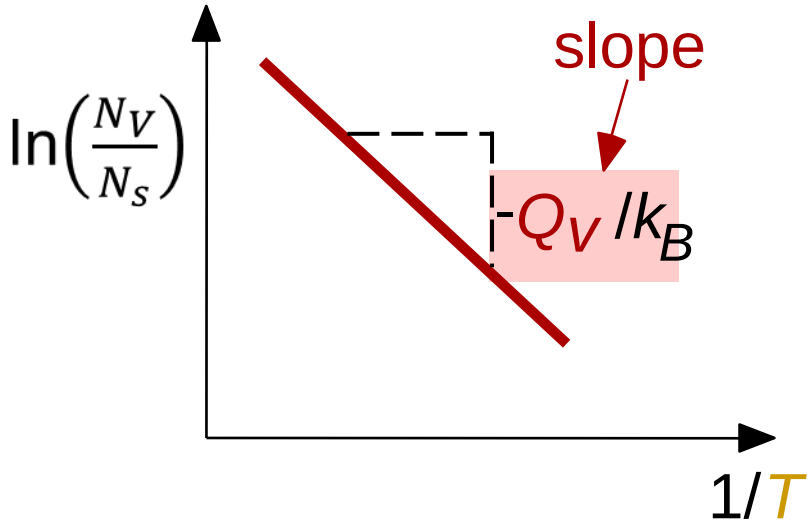
$$\frac{N_V}{N_s} = exp\left(\frac{-Q_V}{k_B T}\right)$$

- Measure this...

$$N_s = \frac{N_A \rho}{A}$$

- Replot it...





Example One- Vacancy Concentrations in Copper

Given that the energy for vacancy formation of Cu at room Temperature, Q_v = 0.9 eV/atom and the density, ρ = 8.4 g/cm^3 . **Estimate the number of vacancies in Cu at room T.** Take Boltzmann's constant $k_B = 1.38\times10^{-23}$ J/atom-K = 8.62×10^{-5} eV/atom-K, $N_A = 6.023 \times 10^{23}$ atoms/mol, and A_{cu} = 63.5 g/mol.

Solution

$$N_v = N_s \exp\left(-Q_v / k_B T\right)$$

- The temperature in Kelvin T = 27°C + 273 = 300 K.
- $k_B T = 300\ K \times 8.62 \times 10^{-5}$ eV/K = 0.026 eV
- The energy for vacancy formation Q_v = 0.9 eV/atom
- The number of regular lattice sites $N_s = N_A \rho / A_{cu}$

Imperfections in solids

Vacancy - examples One Solution

$$N_s = \frac{(6.023 \times 10^{23} atoms/mol) \times (8.4 g/cm^3)}{63.5 g/mol}$$

$$N_s = 8 \times 10^{22} atoms\ or\ sites/cm^3$$

$$N_v = N_s exp\left(-\frac{Q_v}{kT}\right)$$

$$N_v = 8 \times 10^{22} \frac{atoms}{cm^3} exp\left(-\frac{0.9 ev/atom}{0.02 ev/atom}\right)$$

$$N_v = 7.4 \times 10^{7} vacancies/cm^3$$

Example 2
Vacancy Concentrations in Iron

Determine the number of vacancies needed for a BCC iron crystal to have a density of 7.87 g/cm^3. The lattice parameter of the iron is 2.866 × 10^{-8} cm.

Example 2 SOLUTION

The expected theoretical density of iron can be calculated from the lattice parameter and the atomic mass.

$$\rho = \frac{nA}{V_C N_A}$$

Example 2 SOLUTION (Continued)

$$\rho = \frac{nA}{V_C N_A}$$

where

n = number of atoms associated with each unit cell
A = atomic weight
V_C = volume of the unit cell
N_A = Avogadro's number (6.023×10^{23} atoms/mol)

$$\rho = \frac{(2 \text{ atoms/cell})(55.847 \text{ g/mol})}{(2.866 \times 10^{-8} \text{ cm})^3 (6.02 \times 10^{23} \text{ atoms/mol})} = 7.8814 \text{ g/cm}^3$$

Example 2 SOLUTION (Continued)

Let's calculate the number of iron atoms and vacancies that would be present in each unit cell for the required density of 7.87 g/cm^3:

$$\rho = \frac{(X \text{ atoms/cell})(55.847 \text{ g/mol})}{(2.866 \times 10^{-8} \text{ cm})^3 (6.02 \times 10^{23} \text{ atoms/mol})} = 7.87 \text{ g/cm}^3$$

$$X \text{ atoms/cell} = \frac{(7.87)(2.866 \times 10^{-8})^3 (6.02 \times 10^{23})}{55.847} = 1.9971$$

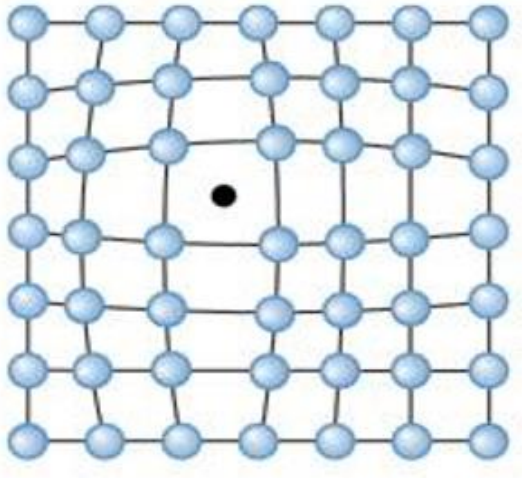
Or, there should be 2.00 – 1.9971 = 0.0029 vacancies per unit cell. The number of vacancies per cm^3 is:

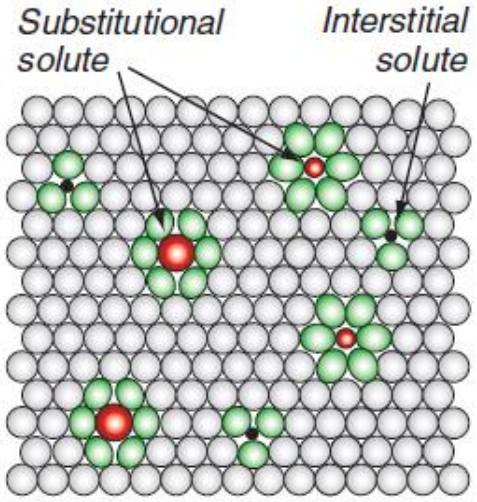
$$\text{Vacancies/cm}^3 = \frac{0.0029 \text{ vacancies/cell}}{(2.866 \times 10^{-8} \text{ cm})^3} = 1.23 \times 10^{20}$$

Interstitial Defects

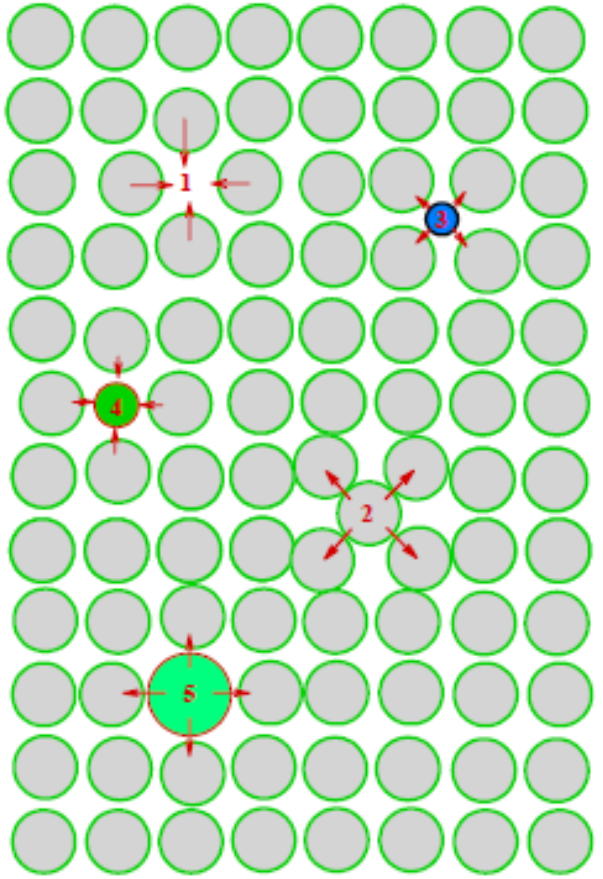
- An Interstitial defect is formed when an extra atom occupies a normally unoccupied site (interstitial site).
- Since the interstitial atoms are larger than the interstitial sites the surrounding lattice is in compression.

- The arrows show the local stresses introduced by the point defects.





interstitial atom



Schematic representation of different point defects:
(1) vacancy;
(2) self-interstitial;
(3) interstitial impurity;
(4,5) substitutional impurities

Self-interstitials

- Self-interstitials in metals introduce large distortions in the surrounding lattice ⇒ the energy of self-interstitial formation is ~ 3 times larger as compared to vacancies ($Q_i \sim 3 \times Q_v$) ⇒ equilibrium concentration of self-interstitials is very low (less than one self-interstitial per cm^3 at room T).

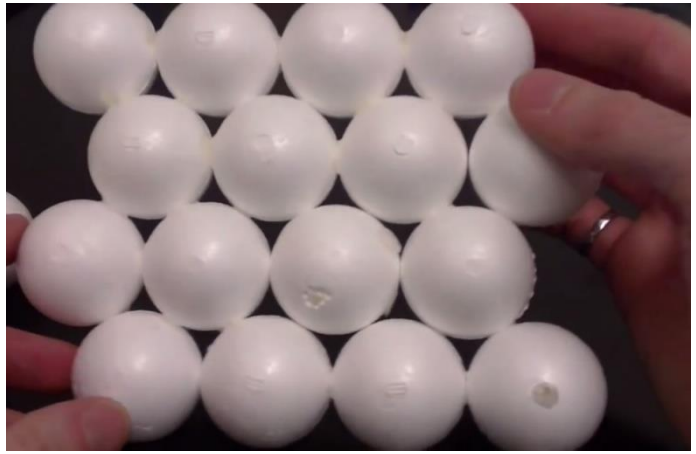
- why is the equilibrium concentration of self-interstitials low?

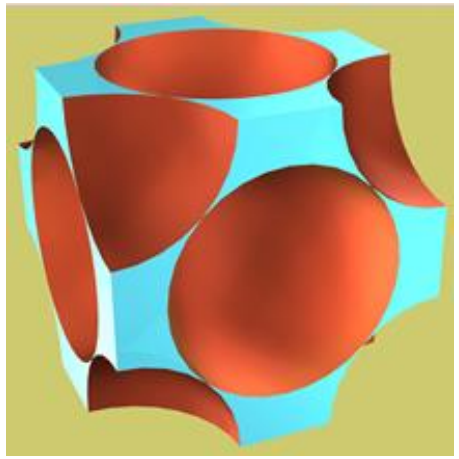
Methods of Producing Point Defects

- Growth and synthesis Impurities may be added to the material during synthesis
- Thermal & thermochemical treatments and other stimuli
  - Heating to high temperature and quench
  - Heating in reactive atmosphere
  - Heating in vacuum ➔e.g. in oxides it may lead to loss of oxygen
  - Etc.
- Plastic Deformation
- Ion implantation and irradiation
- Electron irradiation (typically >1MeV)
  - → Direct momentum transfer or during relaxation of electronic excitations)
  - Ion beam implantation (As, B etc.)
  - Neutron irradiation

Interstitial Sites in Close Packing

- Even in the close packing of spheres, there are some empty spaces between the spheres.
- This empty space in the crystal lattice is called site or void or hole.

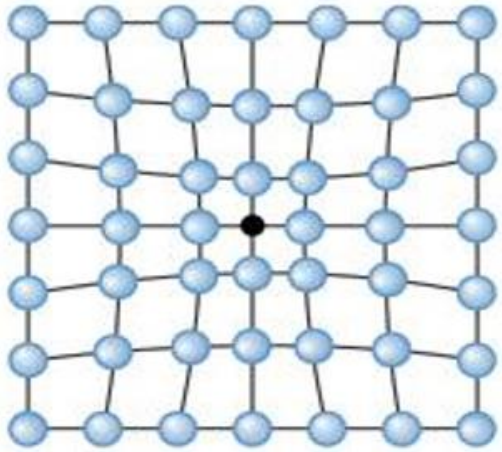




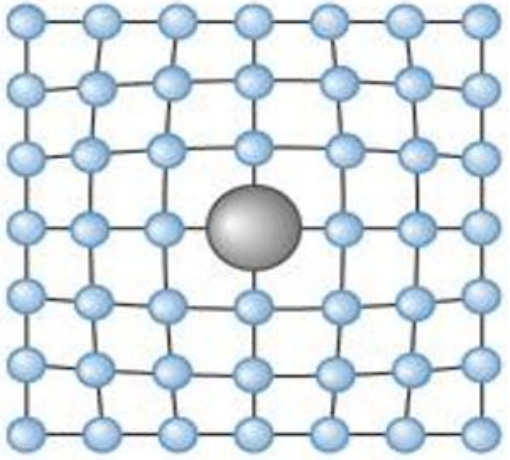
- Generally we can define the void as the unoccupied empty space in **unit cell**. Voids are also known as holes in unit cell.

Substitutional Defects

- Substitutional defects are produced when one atom is replaced by a different type of atom.
- If the substitutional atom is smaller than the original atom then the lattice is in tension.
- If the substitutional atom is larger then the original atom then the lattice is in compression.



(a) small substitutional atom



(b) large substitutional atom

Imperfections in Solids

Conditions for substitutional solid solution (S.S.)

- Hume – Rothery rules
  - 1. Δr (atomic radius) < 15%
  - 2. Proximity in periodic table
    - i.e., similar electronegativities
  - 3. Same crystal structure for pure metals
  - 4. Valency equality
    - All things being equal, a metal will have a greater tendency to dissolve a metal of higher valency than one of lower valency (it provides more electrons to the "cloud")

Imperfections in Solids

Application of Hume–Rothery rules – Solid Solutions

1. Would you predict more Al or Ag to dissolve in Zn?

More Al because size is closer and val. Is higher – but not too much – FCC in HCP

2. More Zn or Al in Cu?

Surely Zn since size is closer thus causing lower distortion (4% vs 12%)

Element	*Atomic Radius (nm)*	*Crystal Structure*	*Electro-nega-tivity*	*Valence*
Cu	0.1278	FCC	1.9	+2
C	0.071			
H	0.046			
O	0.060			
Ag	0.1445	FCC	1.9	+1
Al	0.1431	FCC	1.5	+3
Co	0.1253	HCP	1.8	+2
Cr	0.1249	BCC	1.6	+3
Fe	0.1241	BCC	1.8	+2
Ni	0.1246	FCC	1.8	+2
Pd	0.1376	FCC	2.2	+2
Zn	0.1332	HCP	1.6	+2

Table on p. 106, *Callister 7e.*

Imperfections in Solids

- **Specification of composition**
  - weight percent

$$C_1 = \frac{m_1}{m_1 + m_2} \times 100$$

m_1 = mass of component 1

  - atom percent

$$C_1' = \frac{n_{m1}}{n_{m1} + n_{m2}} \times 100$$

n_{m1} = number of moles of component 1

Impurities

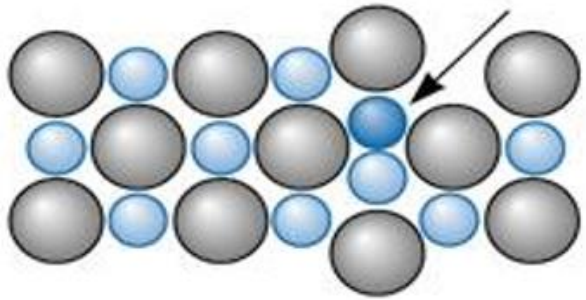
- Impurities - atoms which are different from the host
  - All real solids are impure. Very pure metals 99.9999% - one impurity per 10^6 atoms
  - May be intentional or unintentional

    Examples: carbon added in small amounts to iron makes steel, which is stronger than pure iron. Boron added to silicon change its electrical properties.
  - Alloys - deliberate mixtures of metals

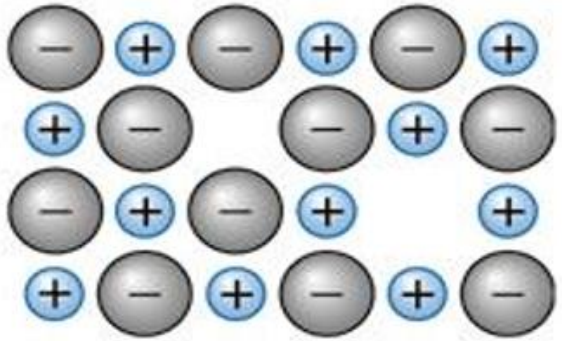
    Example: sterling silver is 92.5% silver – 7.5% copper alloy. Stronger than pure silver.

Other Point Defects

- Frenkel defect - A pair of point defects produced when an ion moves to create an interstitial site, leaving behind a vacancy.
- Schottky defect - A point defect in ionically bonded materials. In order to maintain a neutral charge, a stoichiometric number of cation and anion vacancies must form.



(a) Frenkel defect



(b) Schottky defect

- All of these defects disrupt the perfect arrangement of the surrounding atoms.

Interfacial Defects

External Surfaces

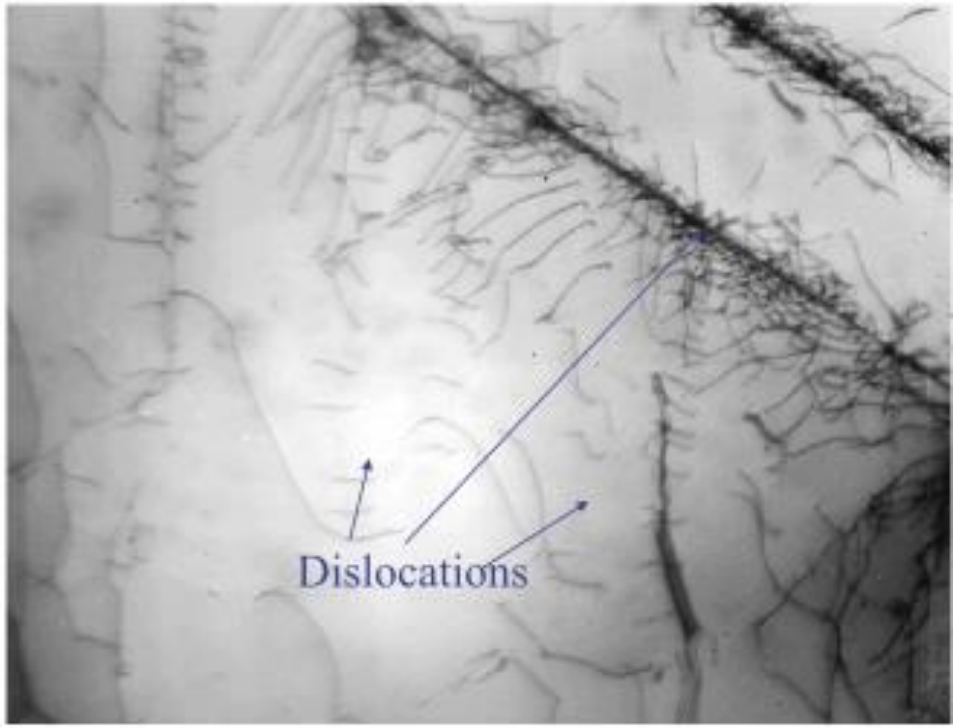
- Surface atoms have unsatisfied atomic bonds, and higher energies than the bulk atoms ⇒ Surface energy, γ (J/m^2)
  - Surface areas tend to minimize (e.g. liquid drop)
  - Solid surfaces can "reconstruct" to satisfy atomic bonds at surfaces.

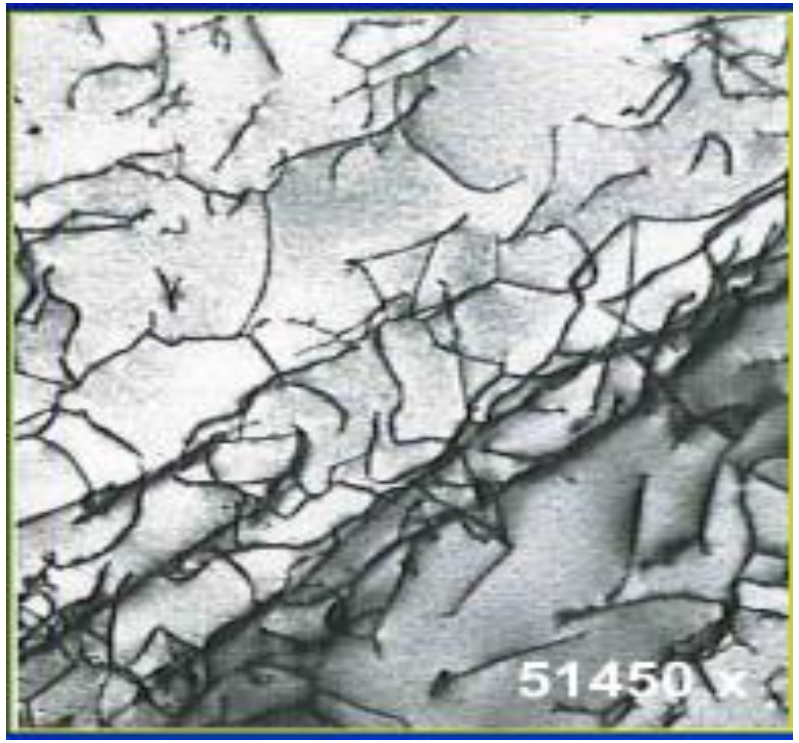
Grain Boundaries

- Polycrystalline material comprised of many small crystals or grains. The grains have different crystallographic orientation. There exist atomic mismatch within the regions where grains meet. These regions are called grain boundaries.
- Surfaces and interfaces are reactive and impurities tend to segregate there. Since energy is associated with interfaces, grains tend to grow in size at the expense of smaller grains to minimize energy. This occurs by diffusion, which is accelerated at high temperatures.

Line Defects: Dislocation in Metals

- Responsible for large mechanical deformation in crystalline solids
- -Linear (one dimensional) defect around which some of the atoms are misaligned





TEM of a Ti alloy
(dark lines are dislocations)

Adapted from Fig. 4.6, *Callister 7e.*

Why Dislocation is Important

- Dislocations play diverse roles in determining materials structures and behaviors.
- The most important role is to weaken the crystal strength.
- The role of dislocations in materials involves the interactions of a dislocation with other dislocations and defects in the material, which result in *'hardening' of the crystal, i.e., strengthening of the weakened crystal.*

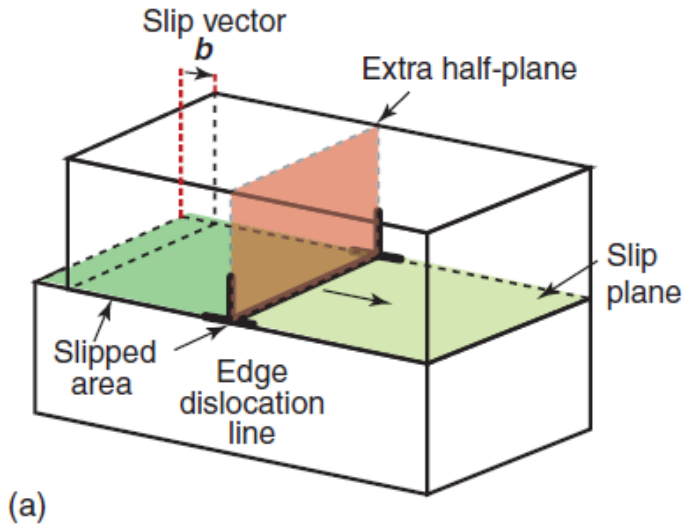
Historical Overview of the Idea of "Dislocations"

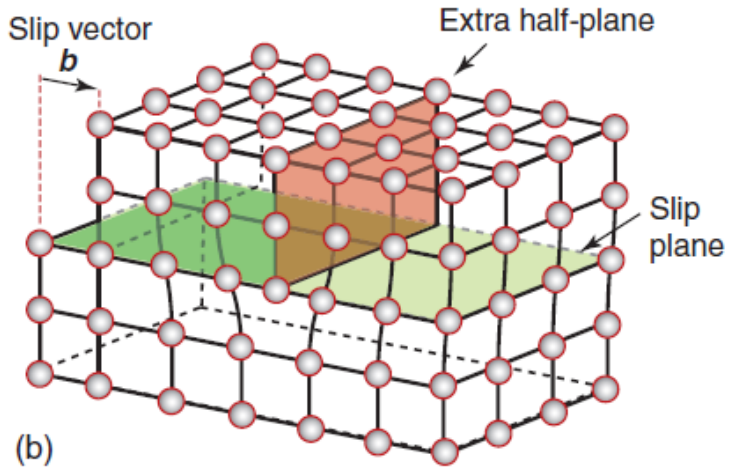
- The continuum construction of a dislocation had been proposed by Volterra in 1905.
- But as late as in 1930, the reason behind the weakening of a crystal was still not clear: *Why a rod of copper can be bent easily.*
- In 1934, Taylor, Orowan and Polanyi postulated the presence of dislocations as a mechanism of weakening of a crystal.
- The presence of dislocations was confirmed by electron microscopy in 1950s

Types of Dislocations

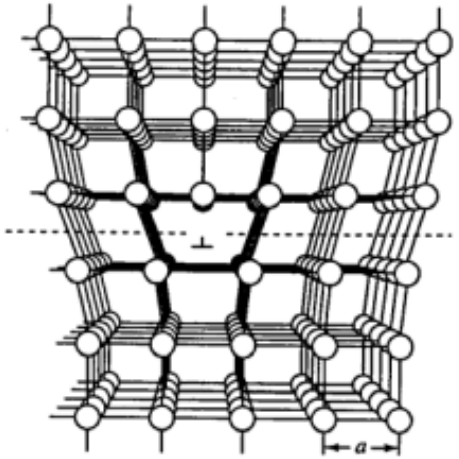
- Edge Dislocation:
  - A portion of an extra plane of atoms
- Screw Dislocation:
  - Helical atomic displacement around a line extending through the crystal
- Mixed Dislocation:
  - Some edge, some screw nature

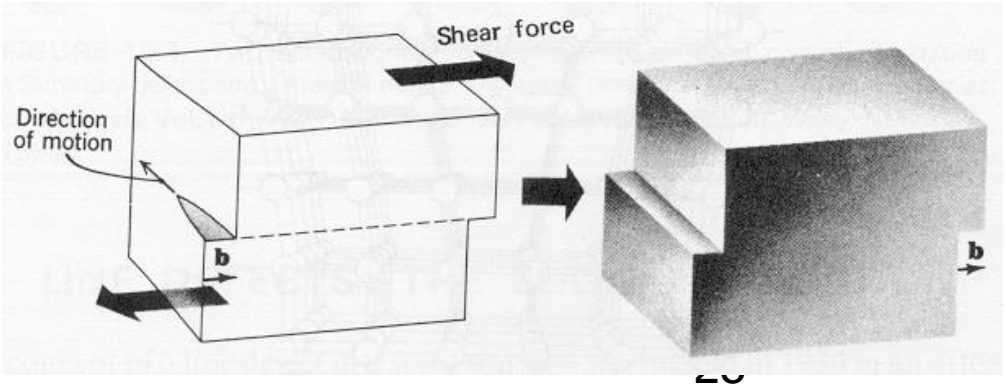
Edge Dislocation





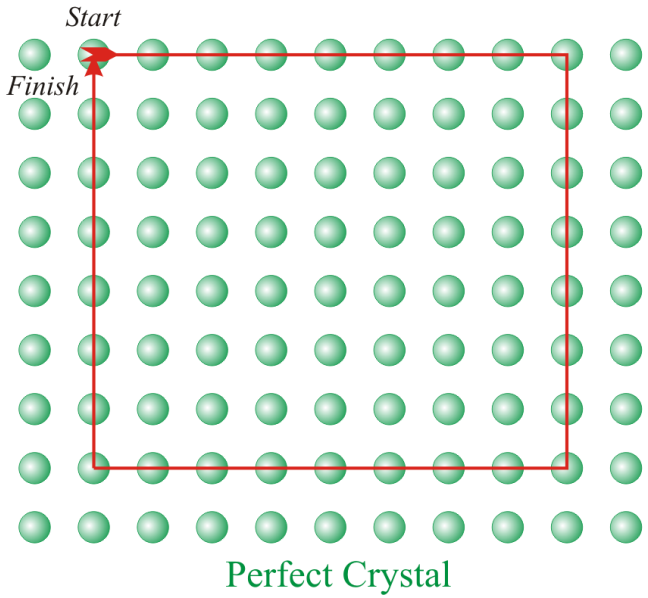
(a) Making a dislocation by cutting, slipping and rejoining bonds across a slip plane. (b) The atom configuration at an edge dislocation in a simple cubic crystal.





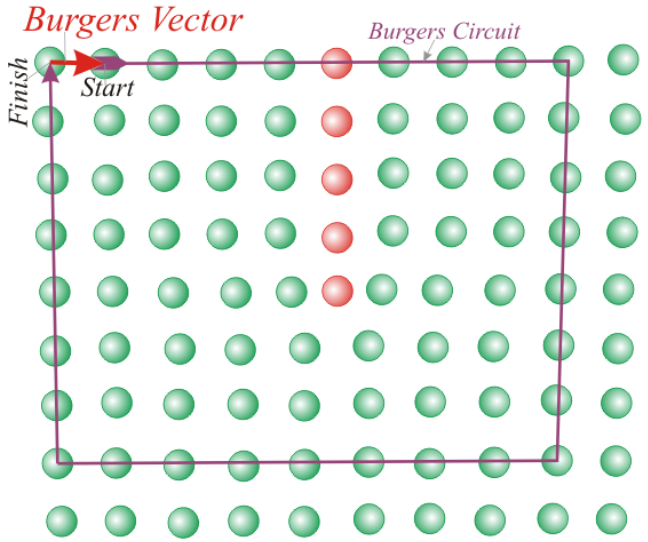
Burgers vector: Edge dislocation

- In a perfect crystal, make a circuit (*e.g.* 8 atomic steps to right, 7 down, 8 left & 7 up). The circuit is *Right Handed*.
- Do the same in the same in the dislocated crystal. The 'missing link' (using some convention like RHFS) is the Burgers vector.



Perfect Crystal

RHFS:
Right Hand Finish to Start convention

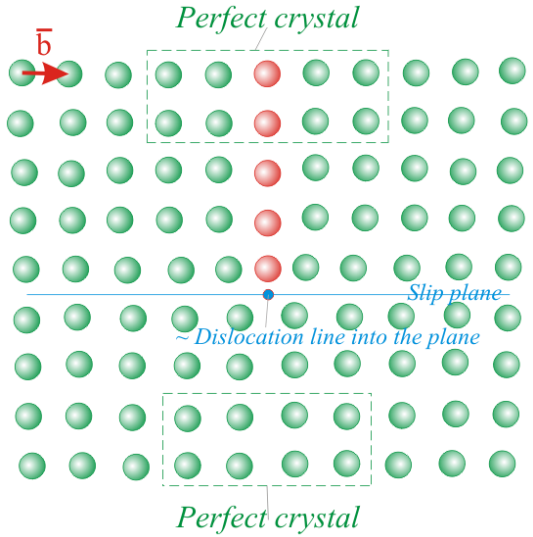


Dislocated Crystal (Edge Dislocation)

Note: the circuit is drawn away from the dislocation line

Understanding the Edge Dislocation

- The edge dislocation is NOT the 'extra half-plane', it is *neither* the 'missing half-plane'
  → it is the line between the 'extra' and the 'missing' half-planes.
- The regions far away from the dislocation line are perfect → all the 'deformation' is concentrated around the dislocation line.
- However, the stress field of the dislocation is a 'long range' field.

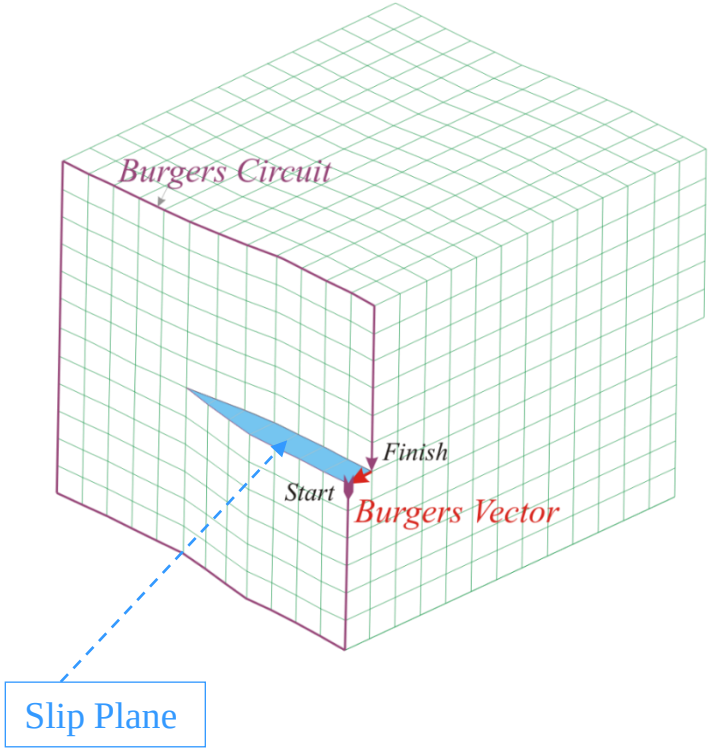


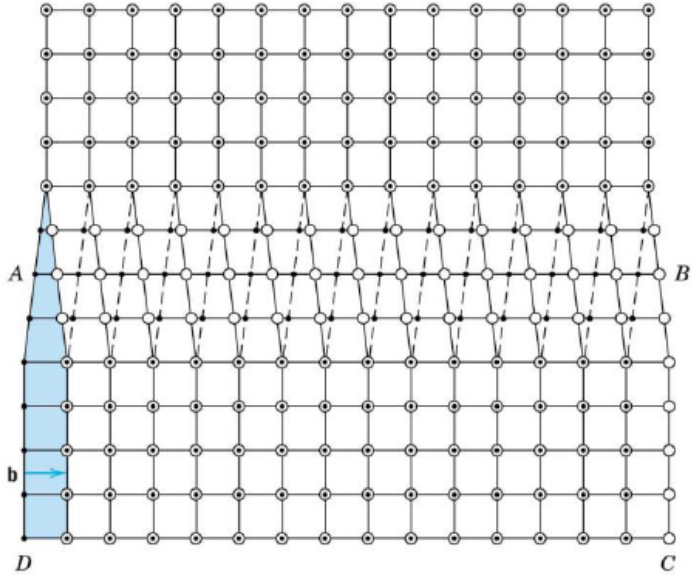
Note:

- The Burgers vector has been drawn away from the dislocation line (sometimes it may be drawn close to dislocation line for convenience).
- The dislocation line is between the 'missing' and 'extra' half-plane.

Screw Dislocation

Burgers vector for a screw dislocation: b || tau (glide plane includes b and any other crystal direction)



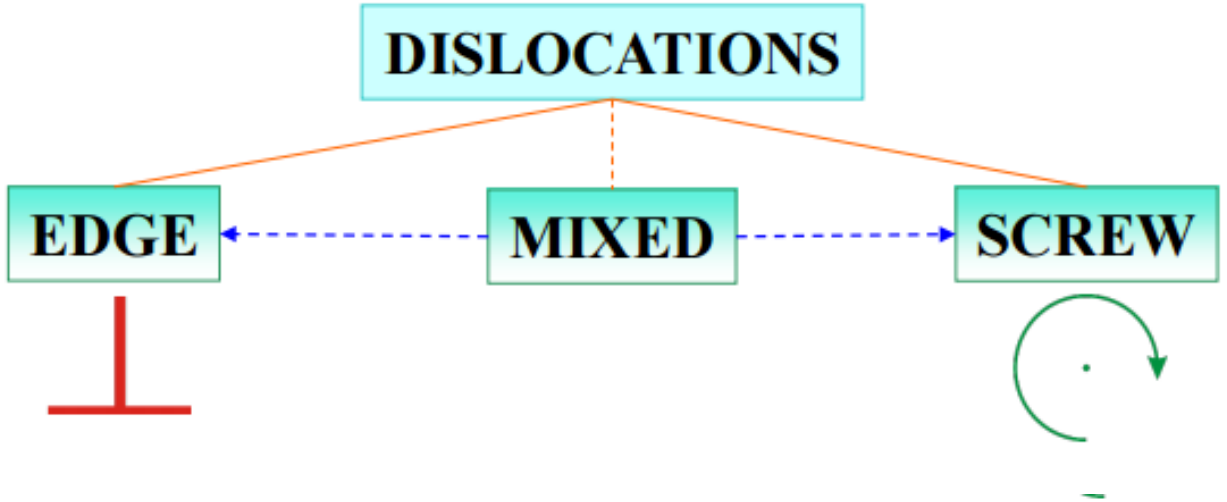


Harder to visualize than edge dislocations

- ❑ Open circles upper half atoms
- ❑ Solid circles lower half atoms
- ❑ Connecting "looks **like a screw**"

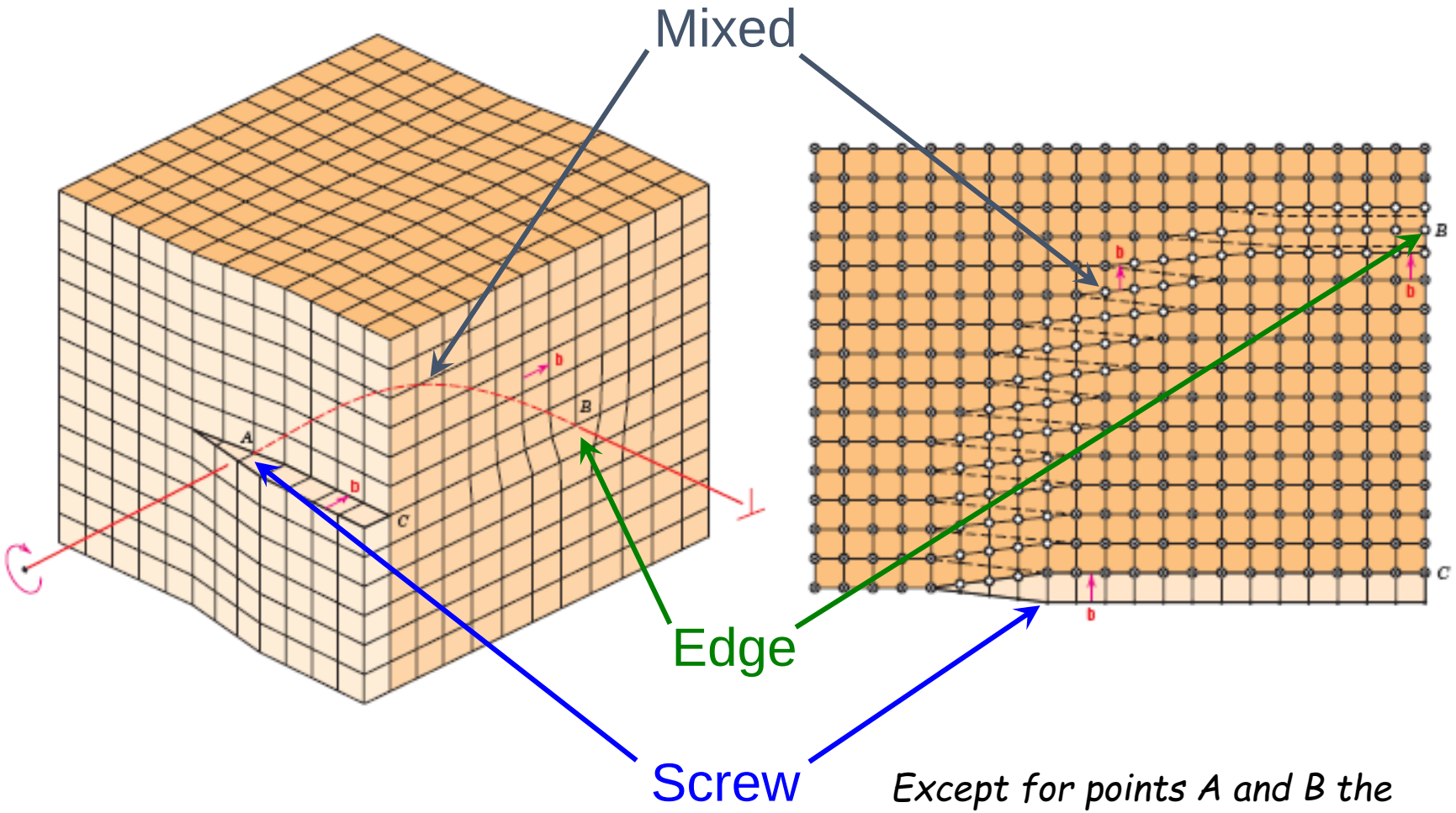
Notes:

The figure shows a Right Handed Screw (RHS) dislocation (RHS is structurally distinct from LHS).
As for the edge dislocation the Burgers circuit has to be drawn far away from the dislocation line.



- Under an observation of Al film with TEM, one usually finds curved dislocation lines, indicating that dislocations have a mixed character and ideal *Edge* and *Screw* dislocations are *extremes*.
- The character of the dislocation will change from position to position along the dislocation line.
- Under special circumstances Pure Edge, Pure Screw or a Mixed Dislocation with a fixed percentage of edge character can form.
  *For example, in a GeSi epitaxial film on Si substrate, 60 ° misfit dislocations can form, where the dislocation lines are straight with the angle between **b** and **t** being 60 °).*

Edge, Screw, and Mixed Dislocations



Adapted from Fig. 4.5, *Callister 7e.*

Except for points A and B the remaining portion of the dislocation line has a mixed character

Edge, Screw, and Mixed Dislocations

Table 1.15 Comparison of Dislocation Characteristics

Dislocation Type	Burger's Vector	Propagation Direction
Edge	$\perp$ to dislocation line	$\parallel$ to dislocation line, $\parallel$ to Burger's vector
Screw	$\parallel$ to dislocation line	$\perp$ to dislocation line, $\perp$ to Burger's vector
Mixed	Neither $\parallel$ nor $\perp$ to dislocation line	Neither $\parallel$ nor $\perp$ to dislocation line, Burger's vector